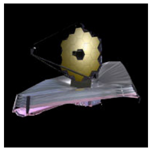

10311 - Slicing the Saguaro: Dissecting a Little Red Dot Descendant at Cosmic Noon with NIRSpec/IFU

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1		NIRSpec IFU Spectroscopy	(1) Saguaro

ABSTRACT

Little Red Dots (LRDs) are compact, extremely red sources abundant at z>4 yet nearly absent by z<4, a sharp decline inconsistent with smooth evolution. Their V-shaped SEDs, overmassive black holes, and elusive hosts challenge current galaxy-formation models. We propose NIRSpec/IFU observations of Saguaro (z=2.1045), the first system combining a canonical LRD-like nucleus with a spatially resolved host, to perform the first spatially and kinematically resolved dissection of an LRD system. Medium- and high-resolution gratings plus PRISM will map rest-frame UV–NIR emission lines and continua at sub-kiloparsec scales, testing whether Saguaro is a true descendant or a low-z analog of high-z LRDs. These data will (1) trace the dynamical and chemical link between the LRD-like nucleus and host; (2) determine whether its black hole remains overmassive or has reached the local MBH–M relation at Cosmic Noon; and (3) spatially resolve the dense nuclear absorber He I 10830 Å observed also in classic LRDs. Whether Saguaro marks the fate of LRDs or their mirror at lower z, these observations will provide the benchmark dataset to interpret LRD evolution and guide next-generation simulations of black-hole–galaxy co-growth.

OBSERVING DESCRIPTION

We will obtain NIRSpec/IFU spectroscopy of Saguaro using three configurations: G140M/F100LP, G235H/F170LP, and G395H/F290LP, plus PRISM for extended UV coverage. This setup provides continuous rest-frame UV–NIR coverage at a resolving power of about R = 1000–2700, allowing us to resolve emission-line kinematics, ionization, and metallicity at sub-kiloparsec scales.

Given the source brightness (mF444W ~ 19 AB) and its isophotal radius, we adopt two IFU pointings with ~50% overlap to maximize empty spaxels for accurate background estimation while ensuring full coverage of both the compact nucleus and the extended host. The medium cycling dither pattern (18) optimizes cosmic-ray rejection and spatial sampling, allowing drizzling to 0.05 per pixel for enhanced resolution.

Blind Target Acquisition is unnecessary: the ~0.1 pointing accuracy easily places the nucleus within the 3×3 IFU field. Each setup includes a dedicated leakcal exposure to correct potential MSA light leakage. The total on-source time is 19.9 hours (plus overheads, totaling 26.7 hours), achieving a signal-to-noise ratio of at least 5 per resolving element across all key diagnostic lines.

This strategy maximizes on-source efficiency, secures robust background subtraction, and ensures uniform spectral and spatial coverage required to disentangle the dynamics, chemical structure, and ionization state of Saguaro and its host.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	Saguaro	RA: 12 36 35.5914 (189.1482975d) Dec: +62 14 24.03 (62.24001d) Equinox: J2000		

Comments:
Category=Galaxy
Description=[Active galactic nuclei, High-redshift galaxies, Infrared galaxies, Quasars]

Observation

Proposal 10311, Observation 1

Fri Mar 13 19:05:03 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

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Template

TA Method	HFF Readout Mode
NONE	false

Mosaic

Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order
2	1	50.0	10.0	0.0	0.0	DEFAULT

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	CYCLING	MEDIUM	1	8	

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G140M/F100LP	NRSIRS2	20	1	false	true	NONE	8	8	11787.823	
2	G140M/F100LP	NRSIRS2	20	1	true	false	NONE	1	1	1473.478	
3	G235H/F170LP	NRSIRS2	20	1	false	true	NONE	8	8	11787.823	
4	G235H/F170LP	NRSIRS2	20	1	true	false	NONE	1	1	1473.478	
5	G395H/F290LP	NRSIRS2	12	1	false	true	NONE	8	8	7119.378	
6	G395H/F290LP	NRSIRS2	12	1	true	false	NONE	1	1	889.922	
7	PRISM/CLEAR	NRSIRS2RAPID	12	1	false	true	NONE	8	8	1517.245	
8	PRISM/CLEAR	NRSIRS2RAPID	12	1	true	false	NONE	1	1	189.656	